Local and Global Quantities for One-Dimensional Linear Elements

ME/BME 736 – Advanced Finite Element Methods

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This handout uses one three-element mesh to explain the distinction between local and global quantities and how element load vectors and stiffness matrices are assembled. The notation is the same as in Chapter 5.

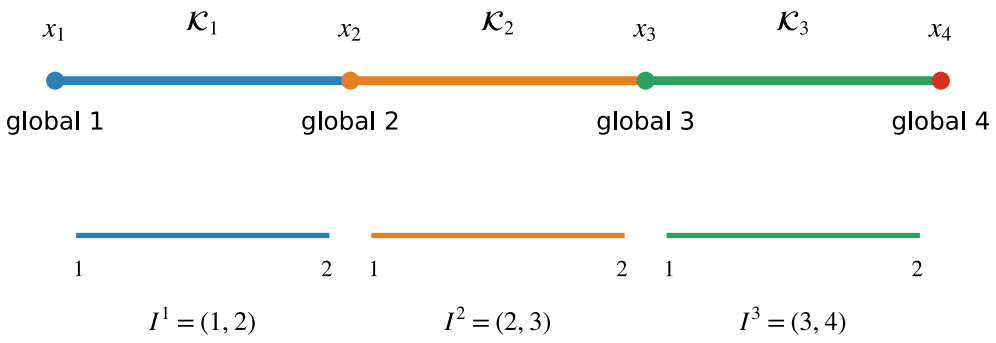
1 One mesh with local and global node numbers

Let

$$0 = x_1 < x_2 < x_3 < x_4 = L$$

be the global nodes of a mesh containing three elements:

$$\mathcal{K}_1 = (x_1, x_2), \qquad \mathcal{K}_2 = (x_2, x_3), \qquad \mathcal{K}_3 = (x_3, x_4).$$



The global node numbers identify nodes in the complete mesh. The local node numbers restart at 1 and 2 on every element. Thus the same physical node x_2 is local node 2 of element 1 and local node 1 of element 2.

1.1 Connectivity and incident elements

The local-to-global map I^e tells us the global number of a local node:

e	$I^e(1)$	$I^e(2)$
1	1	2
2	2	3
3	3	4

$$\Longrightarrow \qquad I^e(1) = e, \quad I^e(2) = e + 1.$$

If global node i belongs to element e, then $(I^e)^{-1}(i)$ gives its local number on that element. For example,

$$(I^1)^{-1}(2) = 2, \qquad (I^2)^{-1}(2) = 1.$$

The superscript is necessary because global node 2 has a different local number in the two elements that meet there.

Finally, let $\mathcal{E}(i)$ denote the elements attached to global node i. For this mesh,

$$\mathcal{E}(1) = \{1\}, \qquad \mathcal{E}(2) = \{1, 2\}, \qquad \mathcal{E}(3) = \{2, 3\}, \qquad \mathcal{E}(4) = \{3\}. \tag{1}$$

Thus

$$\begin{aligned} I^e(a) &: \text{ local node } a \text{ goes to which global node?} \\ (I^e)^{-1}(i) &: \text{ global node } i \text{ has which local number on element } e? \\ \mathcal{E}(i) &: \text{ which elements meet at global node } i? \end{aligned}$$

2 A global finite element function on one element

Let $\{\phi_i\}_{i=1}^4$ be the global nodal basis. The basis function ϕ_i has value one at global node i and zero at every other global node:

$$\phi_i(x_j) = \delta_{ij}. \tag{2}$$

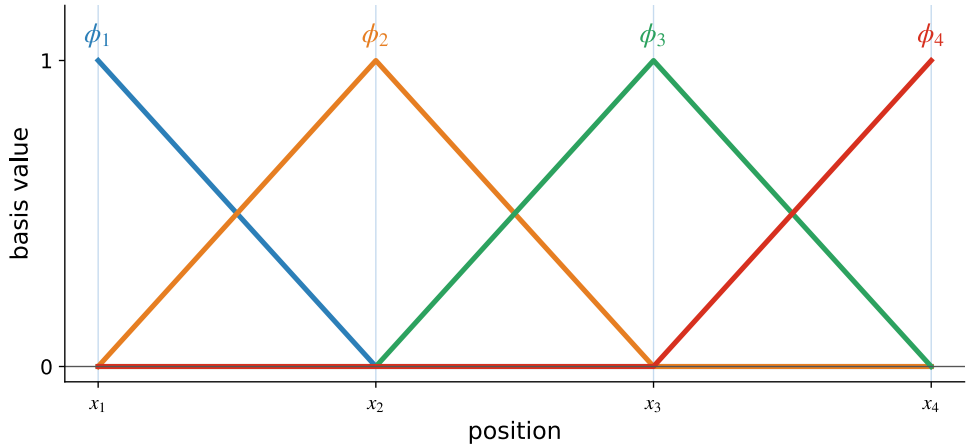
The support of ϕ_i is defined by

$$\operatorname{supp} \phi_i = \overline{\{x \in [0, L] : \phi_i(x) \neq 0\}}. \tag{3}$$

For the linear nodal basis, this support is the union of the closed elements attached to node i:

$$\operatorname{supp} \phi_i = \bigcup_{e \in \mathcal{E}(i)} \overline{\mathcal{K}}_e. \tag{4}$$

Thus an interior basis function is supported on the two elements that meet at its node, while an endpoint basis function is supported on one element.



A finite element function is written globally as

$$u_h(x) = \sum_{i=1}^{4} U_i \phi_i(x). \tag{5}$$

The global degree of freedom U_i is the nodal value $u_h(x_i)$.

On element $\mathcal{K}_e$, only the basis functions associated with its two nodes are nonzero. Number the two local nodes from left to right and define their coordinates by

$$x_1^e = x_e, \qquad x_2^e = x_{e+1}. \tag{6}$$

The corresponding local degrees of freedom are

$$U_1^e = u_h(x_1^e), \qquad U_2^e = u_h(x_2^e). \tag{7}$$

Let ϕ_1^e and ϕ_2^e be the local basis functions:

$$\phi_1^e(x) = \frac{x_{e+1} - x}{h_e}, \qquad \phi_2^e(x) = \frac{x - x_e}{h_e}, \qquad h_e = x_{e+1} - x_e. \tag{8}$$

They satisfy $\phi_a^e(x_b^e) = \delta_{ab}$. Consequently, the restriction of u_h to element e is first written entirely in local quantities as

$$\boxed{u_h|_{\mathcal{K}_e} = U_1^e \phi_1^e + U_2^e \phi_2^e.} \tag{9}$$

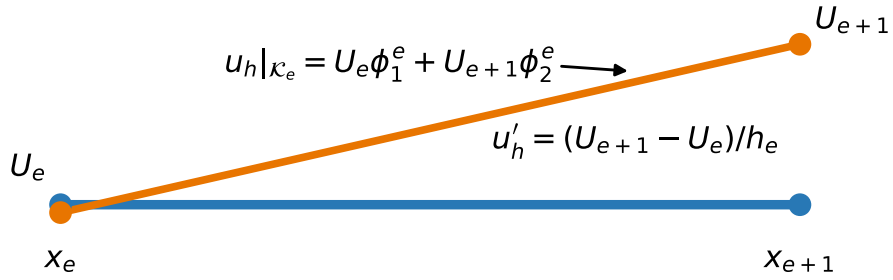
The local-to-global map identifies the same nodes in global notation:

$$U_a^e = U_{I^e(a)}, \qquad \phi_a^e = \phi_{I^e(a)}|_{\mathcal{K}_e}, \qquad a = 1, 2. \tag{10}$$

Substituting these relations into Equation (9) gives

$$\boxed{u_h|_{\mathcal{K}_e} = \sum_{a=1}^{2} U_{I^e(a)} \phi_a^e = U_{I^e(1)} \phi_1^e + U_{I^e(2)} \phi_2^e.} \tag{11}$$

For the consecutive numbering used here, $I^e(1) = e$ and $I^e(2) = e + 1$, so the two coefficients in Equation (11) are U_e and U_{e+1}.



For example, $I^2 = (2, 3)$, and therefore

$$u_h|_{\mathcal{K}_2} = U_2 \phi_1^2 + U_3 \phi_2^2, \qquad \boldsymbol{U}^2 = \begin{bmatrix} U_2 \\ U_3 \end{bmatrix}. \tag{12}$$

For all three elements, the local coefficient vectors are

$$\boldsymbol{U}^1 = \begin{bmatrix} U_1 \\ U_2 \end{bmatrix}, \qquad \boldsymbol{U}^2 = \begin{bmatrix} U_2 \\ U_3 \end{bmatrix}, \qquad \boldsymbol{U}^3 = \begin{bmatrix} U_3 \\ U_4 \end{bmatrix}.$$

3 Assembly of the load vector

For a distributed load f, the global load entry is

$$F_i = \int_0^L f \phi_i \, \mathrm{d}x. \tag{13}$$

Splitting the integral over all three elements first gives

$$F_i = \sum_{e=1}^{3} \int_{\mathcal{K}_e} f\phi_i \, \mathrm{d}x. \tag{14}$$

If element e does not belong to $\mathcal{E}(i)$, then $\phi_i|_{\mathcal{K}_e} = 0$, and its integral in Equation (14) is zero. The full sum therefore reduces to

$$F_i = \sum_{e\in\mathcal{E}(i)} \int_{\mathcal{K}_e} f\phi_i \, \mathrm{d}x. \tag{15}$$

Define the two local load entries on element e by

$$F_a^e = \int_{\mathcal{K}_e} f\phi_a^e \, \mathrm{d}x, \qquad a = 1, 2. \tag{16}$$

Because $\phi_i|_{\mathcal{K}_e} = \phi^e_{(I^e)^{-1}(i)}$, Equation (15) becomes

$$\boxed{F_i = \sum_{e\in\mathcal{E}(i)} F^e_{(I^e)^{-1}(i)}.} \tag{17}$$

For the three-element mesh,

$$F_1 = F_1^1, \qquad F_2 = F_2^1 + F_1^2, \qquad F_3 = F_2^2 + F_1^3, \qquad F_4 = F_2^3.$$

Thus the global load vector is the sum of the three element vectors placed in their global positions:

$$\boxed{\boldsymbol{F} = \begin{bmatrix} F_1^1 \\ F_2^1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ F_1^2 \\ F_2^2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ F_1^3 \\ F_2^3 \end{bmatrix} = \begin{bmatrix} F_1^1 \\ F_2^1 + F_1^2 \\ F_2^2 + F_1^3 \\ F_2^3 \end{bmatrix}.} \tag{18}$$

At a shared node, the contributions from the neighboring elements are added.

4 Assembly of the stiffness matrix

The global stiffness entry for the axial bar is

$$K_{ij} = \int_0^L AE \, \phi_j' \phi_i' \, \mathrm{d}x. \tag{19}$$

Define the local stiffness entries on element e by

$$K_{ab}^e = \int_{\mathcal{K}_e} AE \, (\phi_b^e)'(\phi_a^e)' \, \mathrm{d}x, \qquad a, b = 1, 2. \tag{20}$$

An element contributes to K_{ij} only when it contains both global nodes i and j. Hence

$$\boxed{K_{ij} = \sum_{e\in\mathcal{E}(i)\cap\mathcal{E}(j)} K^e_{(I^e)^{-1}(i),(I^e)^{-1}(j)}.} \tag{21}$$

For example,

$$K_{22} = K_{22}^1 + K_{11}^2, \qquad K_{23} = K_{12}^2, \qquad K_{14} = 0.$$

Let

$$\boldsymbol{K}^e = \begin{bmatrix} K^e_{11} & K^e_{12} \\ K^e_{21} & K^e_{22} \end{bmatrix}.$$

Placing each 2×2 element matrix into the rows and columns specified by I^e gives

$$\begin{aligned} \boldsymbol{K} &= \begin{bmatrix} K^1_{11} & K^1_{12} & 0 & 0 \\ K^1_{21} & K^1_{22} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & K^2_{11} & K^2_{12} & 0 \\ 0 & K^2_{21} & K^2_{22} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ &\quad + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & K^3_{11} & K^3_{12} \\ 0 & 0 & K^3_{21} & K^3_{22} \end{bmatrix} \\ &= \begin{bmatrix} K^1_{11} & K^1_{12} & 0 & 0 \\ K^1_{21} & K^1_{22} + K^2_{11} & K^2_{12} & 0 \\ 0 & K^2_{21} & K^2_{22} + K^3_{11} & K^3_{12} \\ 0 & 0 & K^3_{21} & K^3_{22} \end{bmatrix}. \end{aligned} \tag{22}$$

For a linear bar element with constant AE,

$$\boldsymbol{K}^e = \frac{(AE)_e}{h_e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \tag{23}$$

If $k_e = (AE)_e/h_e$, the assembled matrix is

$$\boldsymbol{K} = \begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1 + k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2 + k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix}. \tag{24}$$

5 The assembly rule

For every element e, compute its two local load entries and its four local stiffness entries. Then use I^e to add them to the global vector and matrix:

$$\boxed{F_{I^e(a)} \mathrel{+}= F^e_a, \qquad K_{I^e(a),I^e(b)} \mathrel{+}= K^e_{ab}.} \tag{25}$$

The corresponding algorithm is

```
initialize F = 0 and K = 0
for each element e = 1,...,N_e:
    compute F^e and K^e
    for a = 1,2:
        i = I^e(a)
        F_i += F_a^e
        for b = 1,2:
            j = I^e(b)
            K_ij += K_ab^e
```

In Python, subtract one from every mathematical node and element number when using an array index. The mathematical derivation itself remains one-based.